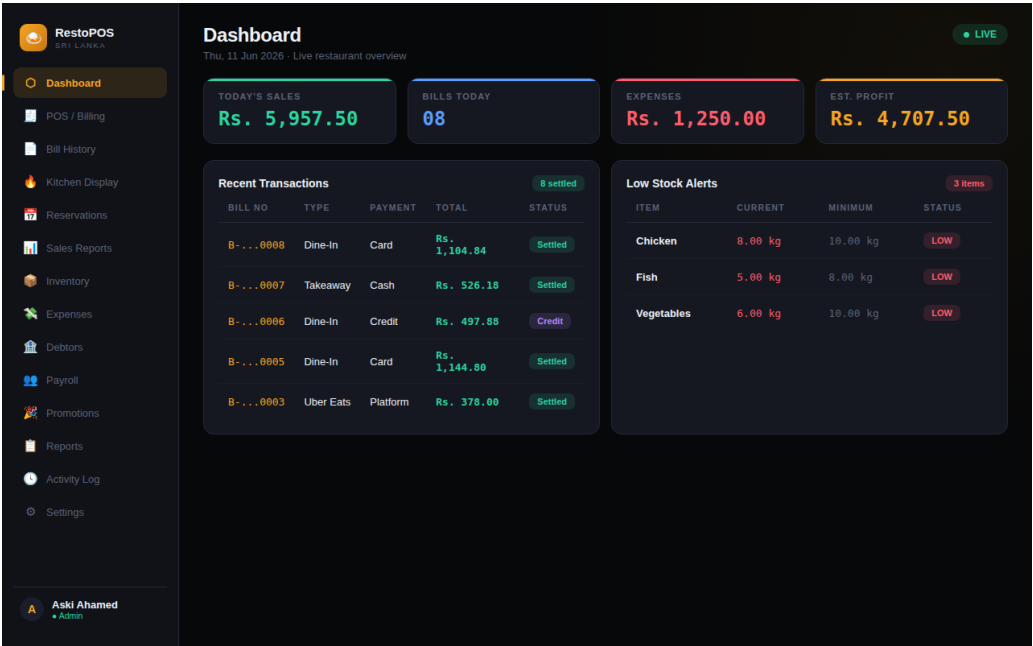


RESTOPOS SRI LANKA

A Web-Based Restaurant Management System for the Sri Lankan Hospitality Industry

Project Documentation Report



Submitted by	System	Year
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ABSTRACT

RestoPOS Sri Lanka is a comprehensive, web-based restaurant management system designed and developed to address the operational needs of small to medium-sized restaurants in Sri Lanka. The system integrates point-of-sale billing, inventory management, employee payroll compliant with Sri Lankan EPF/ETF statutory requirements, customer relationship features such as debtor credit accounts and table reservations, and real-time kitchen order coordination through a dedicated Kitchen Display System.

The system was built using PHP and MySQL on the server side, with a responsive HTML, CSS, and vanilla JavaScript front end, requiring no external frameworks. A role-based access control model restricts functionality across four distinct user roles — Administrator, Manager, Cashier, and Kitchen Staff — ensuring that each user interacts only with the modules relevant to their responsibilities. The application further supports promotional discount campaigns, multi-channel order tracking across dine-in, takeaway, and third-party delivery platforms (Uber Eats, PickMe), and a complete suite of financial reports exportable to PDF and CSV formats.

This document presents the complete system design, module architecture, database schema, and a detailed walkthrough of every functional component, supported by annotated screenshots captured directly from the deployed application.

Keywords: Point of Sale, Restaurant Management System, PHP, MySQL, Role-Based Access Control, Kitchen Display System, EPF/ETF Payroll, Sri Lanka

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CHAPTER 1

Introduction

1.1 Background

The restaurant industry in Sri Lanka has experienced steady growth alongside the expansion of food delivery platforms such as Uber Eats and PickMe. Despite this growth, many small and medium-sized restaurants continue to rely on manual record-keeping, disconnected spreadsheets, or generic point-of-sale software not tailored to local regulatory requirements such as the Employees' Provident Fund (EPF) and Employees' Trust Fund (ETF) statutory deductions.

RestoPOS Sri Lanka was conceived to fill this gap — a purpose-built, locally relevant restaurant management system that handles the full operational lifecycle of a restaurant: from taking a customer's order at the point of sale, to coordinating that order with the kitchen, to tracking inventory consumption, to generating statutory-compliant payroll for staff.

1.2 Problem Statement

Restaurants operating without an integrated management system typically face the following challenges:

- Billing errors and inconsistent service charge or tax application across different order channels (dine-in, takeaway, delivery).
- No reliable way to track which menu items are running low in stock, leading to wasted customer time or lost sales.
- Manual payroll calculation prone to error, particularly for Sri Lanka's EPF (8% employee / 12% employer) and ETF (3% employer) statutory contributions.
- Miscommunication between front-of-house staff and the kitchen, resulting in delayed or incorrect order preparation.
- Lack of audit visibility — no record of who voided a bill, changed a price, or modified a reservation.
- No centralized system for managing promotional campaigns or tracking their actual cost to the business.

1.3 Objectives

The primary objectives of this project were defined as follows:

1. Design and implement a unified point-of-sale system supporting multiple order types (Dine-In, Takeaway, Uber Eats, PickMe, Delivery) with automatic service charge and tax calculation.
2. Develop a real-time Kitchen Display System (KDS) to digitally route orders from the POS directly to kitchen staff, eliminating paper tickets.

3. Build a Sri Lanka-compliant payroll module that automatically calculates EPF and ETF contributions and overtime pay based on actual clock-in/clock-out attendance data.
4. Implement role-based access control to ensure Administrators, Managers, Cashiers, and Kitchen Staff each see only the functionality relevant to their role.
5. Provide a complete suite of exportable financial reports (Profit & Loss, Cash Flow, Inventory, Payroll, Promotions) to support business decision-making.
6. Introduce an audit-trail Activity Log so that all significant user actions are traceable for accountability.
7. Deliver a table reservation system that allows front-of-house staff to record, search, and print booking confirmations.

CHAPTER 2

System Overview

2.1 Technology Stack

The system was implemented using a traditional, dependency-light server-rendered architecture, chosen for ease of deployment on standard shared hosting environments common among small Sri Lankan businesses.

Server-Side Language	PHP 8.0+
Database	MySQL / MariaDB (PDO with prepared statements)
Front-End	HTML5, CSS3, vanilla JavaScript (no external framework)
Typography	Space Grotesk (UI), JetBrains Mono (numeric/monospace data)
Deployment Target	Shared hosting (cPanel) or local XAMPP environment
Authentication	Session-based, role-stored in PHP session
Document Generation	Server-rendered HTML to PDF via browser print dialog
Architecture Pattern	Procedural MVC-inspired — modules per feature, shared includes for config/auth/layout

2.2 User Roles & Access Control

Access to the system is governed by a centralized permission map. Each user account is assigned exactly one role, and the application enforces access restrictions at three independent layers: navigation visibility, direct page access, and data export endpoints.

Administrator	Full access to all 17 modules, including system Settings and user management.
Manager	Full operational access — POS, reports, inventory, payroll, employees — excluding system Settings.
Cashier	Operational subset: Dashboard, POS, Bill History, Sales Reports, Inventory, Expenses, Debtors, limited Rep
Kitchen Staff	Restricted to the Kitchen Display System only. Login redirects directly to the kitchen queue; all other modules

CHAPTER 3

System Modules

This chapter documents each functional module of RestoPOS Sri Lanka in detail, supported by screenshots captured directly from the live application. Each section describes the module's purpose, its key interface elements, and the underlying business logic.

3.1 Authentication & Login

The system uses a session-based authentication mechanism. Upon successful login, the user's role is stored in the PHP session and used by every subsequent page request to determine accessible functionality. Following authentication, the user is redirected to the first module their role is permitted to access — for example, Administrators land on the Dashboard, while Kitchen Staff are routed directly to the Kitchen Display System.

3.2 Dashboard

The Dashboard serves as the system's landing page for management roles, providing a real-time operational snapshot. It aggregates live data from the bills, expenses, and inventory tables without requiring manual refresh of underlying queries.

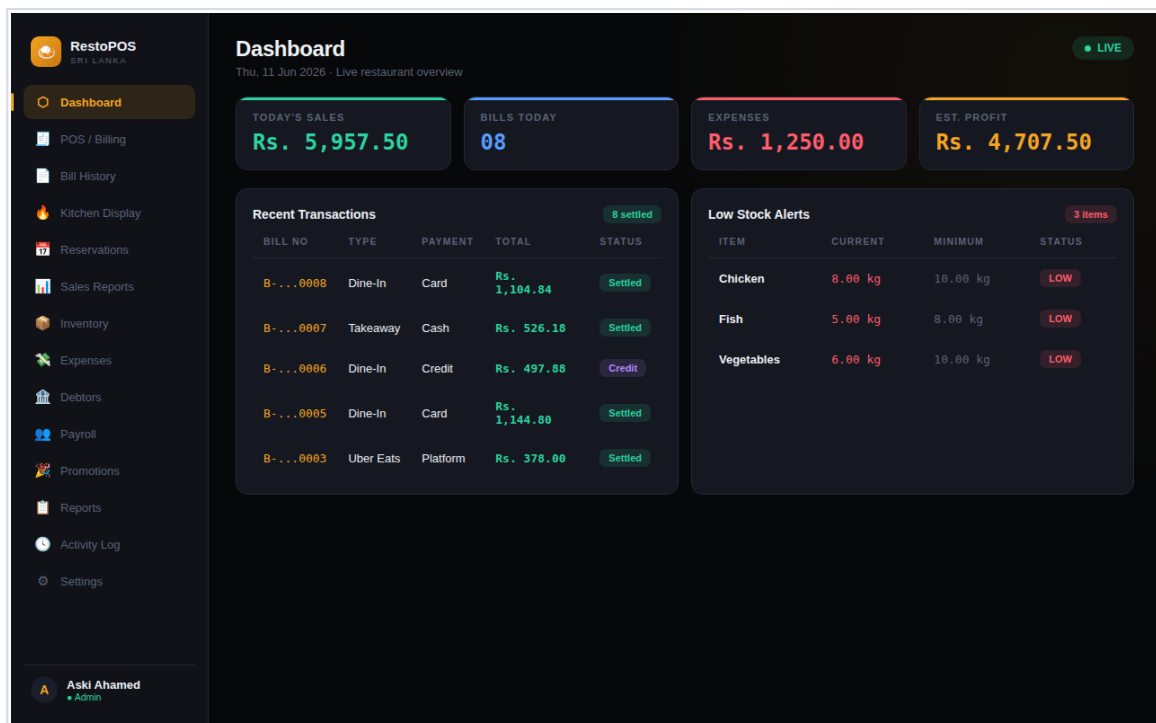


Figure 1: Dashboard showing live sales, bill count, expenses, and low-stock alerts

Key components include: **Today's Sales** (sum of settled bill totals for the current calendar date), **Today's Bills** (transaction count), **Expenses** (recorded outgoings for the day), **Estimated Profit** (sales minus expenses), and a **Low Stock Alerts** panel that highlights any inventory item whose

quantity has fallen to or below its configured minimum threshold.

3.3 Point of Sale (POS) Billing

The POS module is the operational core of the system. It presents a searchable, category-filtered menu grid alongside a live-updating bill panel. The system supports five distinct order types — Dine-In, Takeaway, Uber Eats, PickMe, and Delivery — each with different downstream calculation rules; notably, the 10% service charge applies exclusively to Dine-In orders.

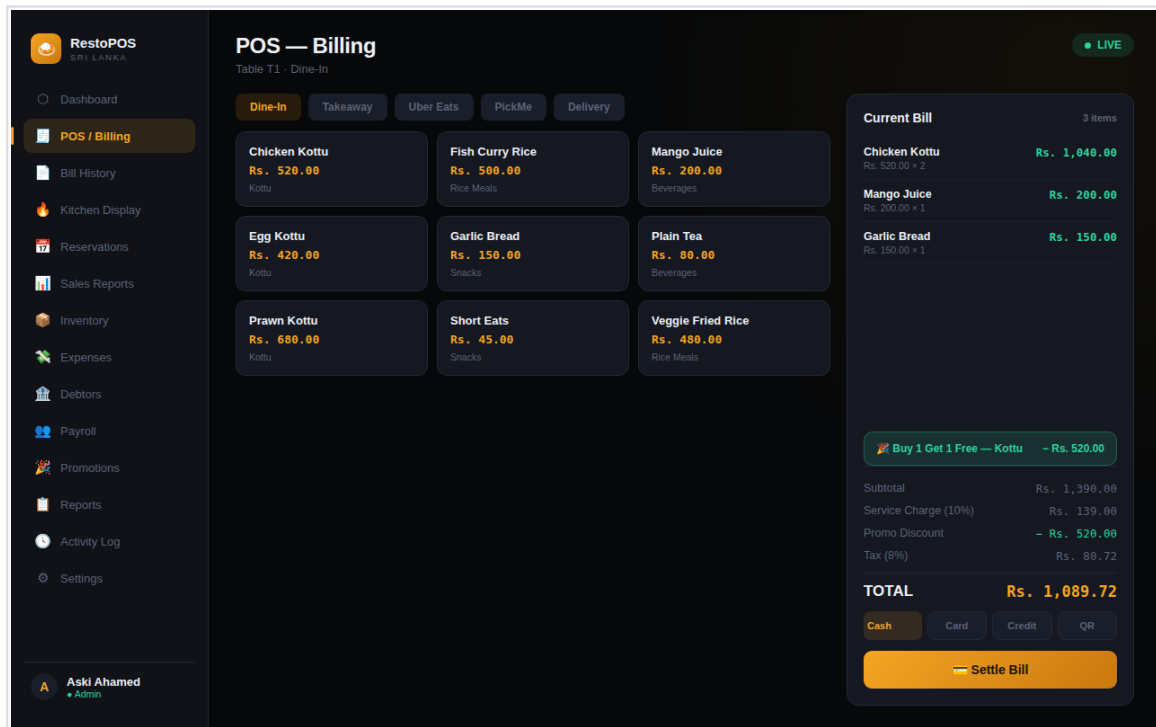


Figure 2: POS billing interface with live cart, promotion application, and payment selection

All monetary calculations — subtotal, service charge, discount, promotional discount, tax, and final total — are computed by a single authoritative client-side function that re-executes on every cart modification. This design was specifically adopted to eliminate calculation drift between multiple competing logic paths, a defect identified and resolved during development (see Chapter 5, Testing & Validation).

Upon settlement, the system generates a unique bill number, persists the transaction and its line items to the database, and immediately offers a printable, auto-launching 80mm thermal receipt.

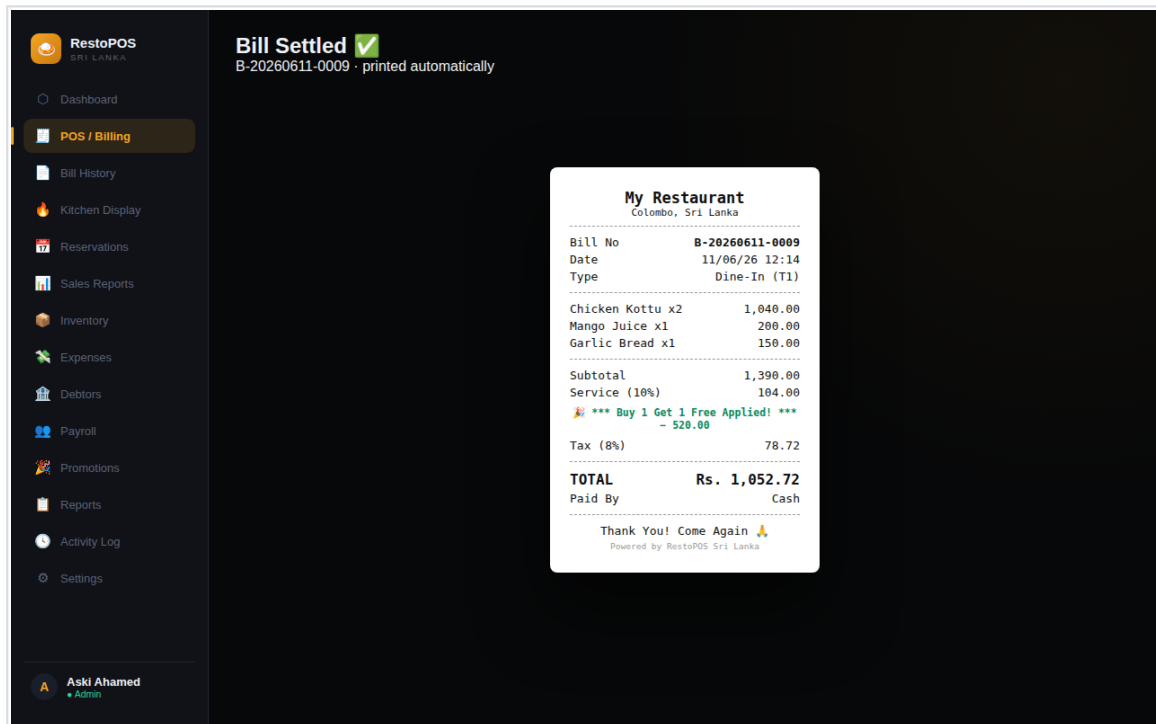


Figure 3: Auto-printing 80mm thermal receipt, including an applied promotional discount line

3.4 Kitchen Display System

The Kitchen Display System (KDS) digitally replaces paper kitchen tickets. Every item added to a settled bill is automatically assigned a kitchen status of "pending." The KDS groups items by their parent bill and presents them as cards, colour-coded by aggregate status (pending, in preparation, or ready).

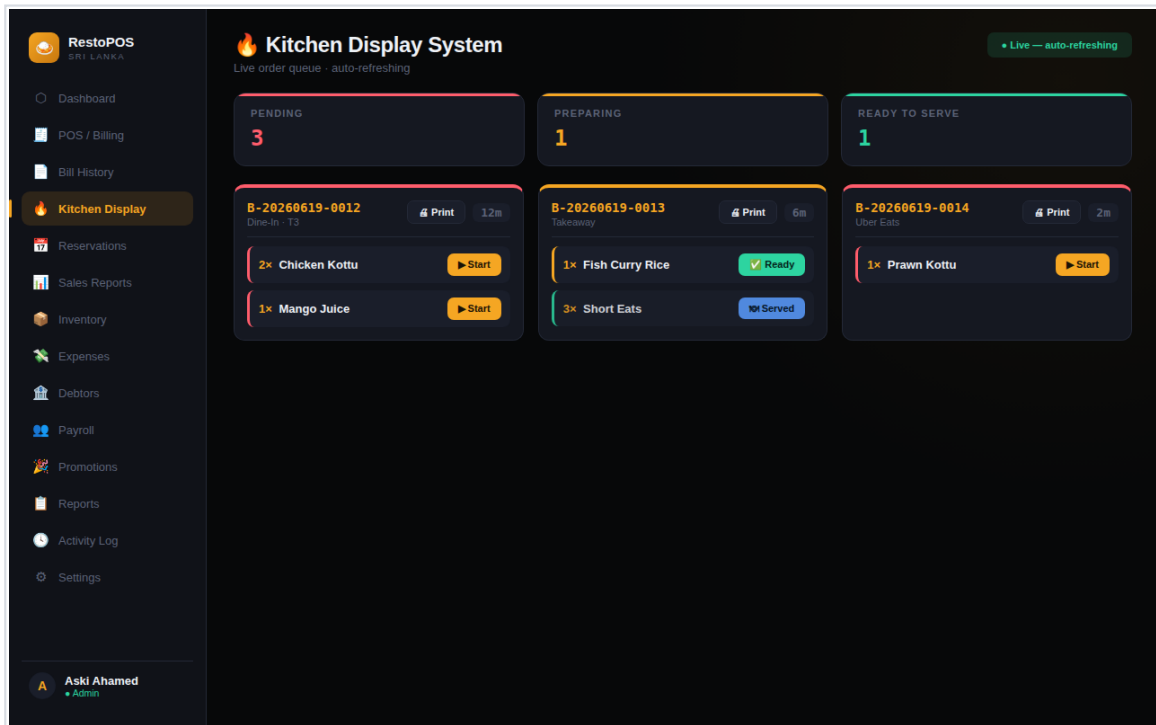


Figure 4: Kitchen Display System showing live order queue grouped by bill, with status-progression controls

Kitchen staff progress each item through a four-stage workflow — *pending* → *preparing* → *ready* → *served* — via single-tap buttons that submit asynchronous (AJAX) status updates without a full page reload. The display automatically refreshes every twenty seconds to surface newly settled orders. Each ticket includes a dedicated **Print** action that generates a compact, thermal-printer-formatted Kitchen Order Ticket (KOT) listing only item names and quantities, omitting pricing information not relevant to kitchen operations.

3.5 Bill History

Bill History provides a chronological, filterable record of every transaction processed through the POS. Administrators and Managers may void or permanently delete erroneous bills; Cashiers retain view and reprint access only, preserving transactional integrity at the point of sale.

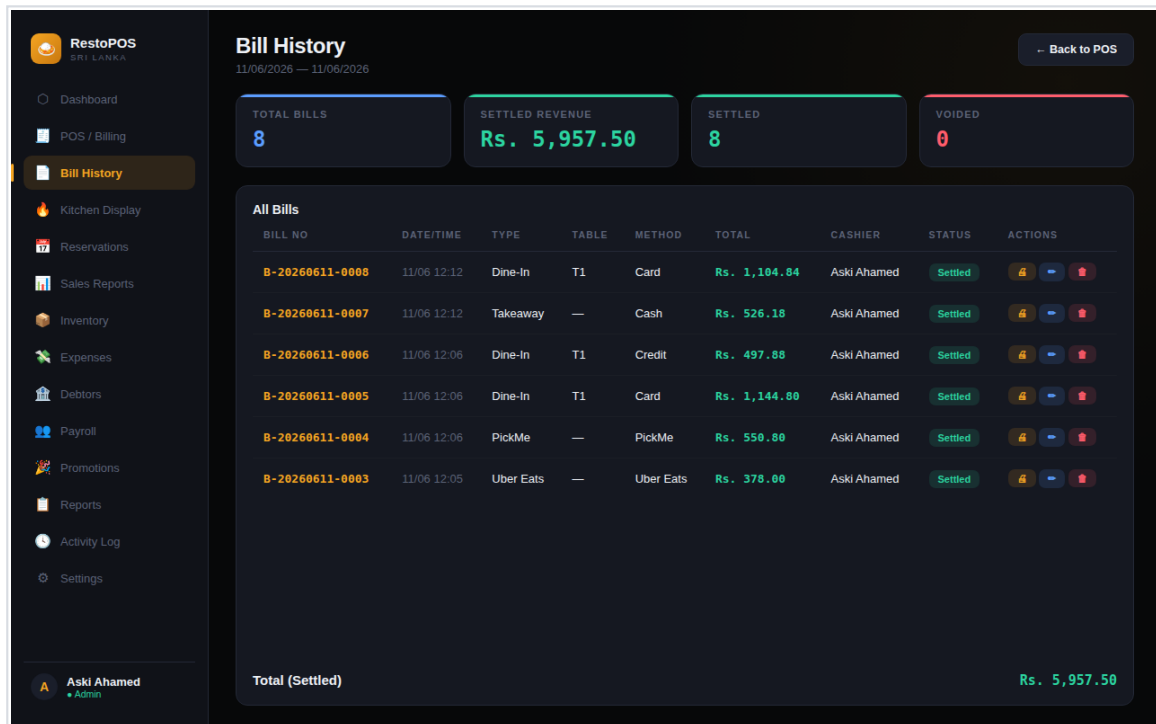


Figure 5: Bill History with date-range filtering and per-row print, edit, void, and delete actions

3.6 Table Reservations

The Reservations module allows front-of-house staff to record customer bookings, capturing customer name, contact number, reservation date and time, party size (pax), preferred location or table, and free-text notes for special requests. An instant client-side search filters the reservation list in real time as the user types, without triggering a server round-trip.

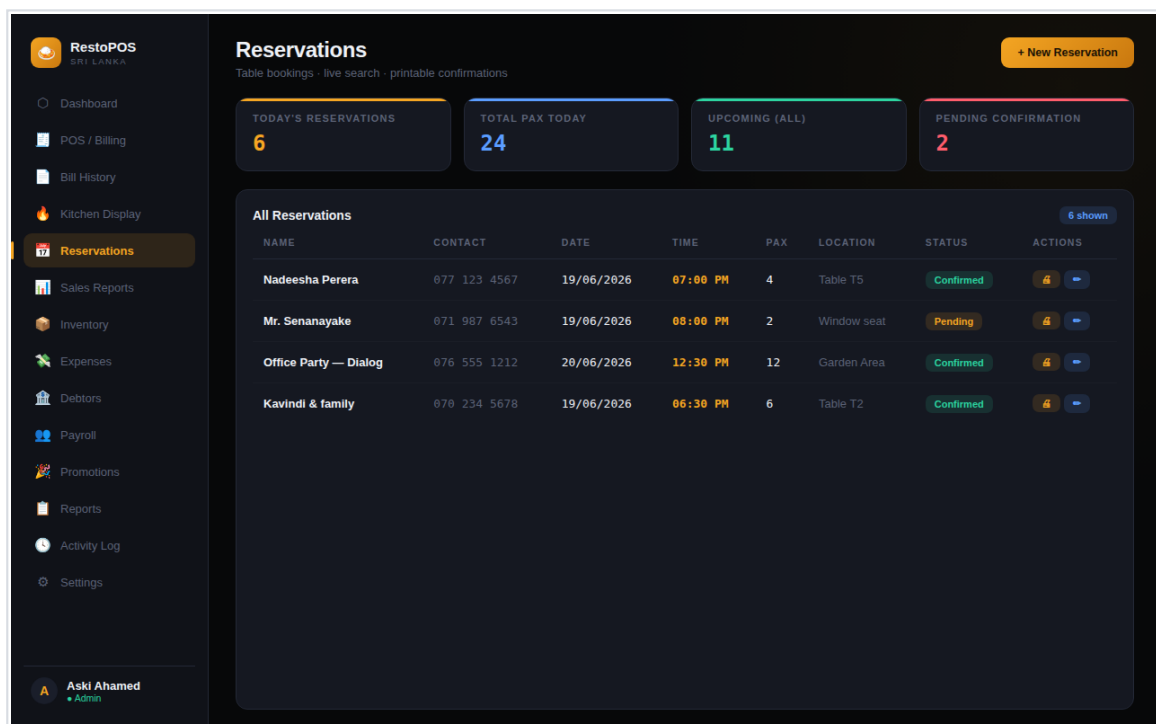


Figure 6: Reservations module with live search and per-row status management

Each reservation can be exported as a standalone, print-ready confirmation document, formatted for either internal record-keeping or as a customer-facing booking confirmation.

3.7 Sales Reports

The Sales Reports module aggregates transactional data across a configurable date range, presenting revenue broken down by order channel and payment method, a top-selling-items ranking, and a complete transaction listing.



Figure 7: Sales Reports showing channel breakdown and top-selling menu items

3.8 Inventory Management

Inventory tracking supports a comprehensive range of units of measurement — including kilograms, grams, milligrams, litres, millilitres, packets, boxes, dozens, bottles, and cylinders — to accommodate the diverse range of ingredients and supplies used in restaurant operations. Each item carries a configurable minimum-quantity threshold; items at or below this threshold are automatically flagged across the Dashboard and Inventory views.

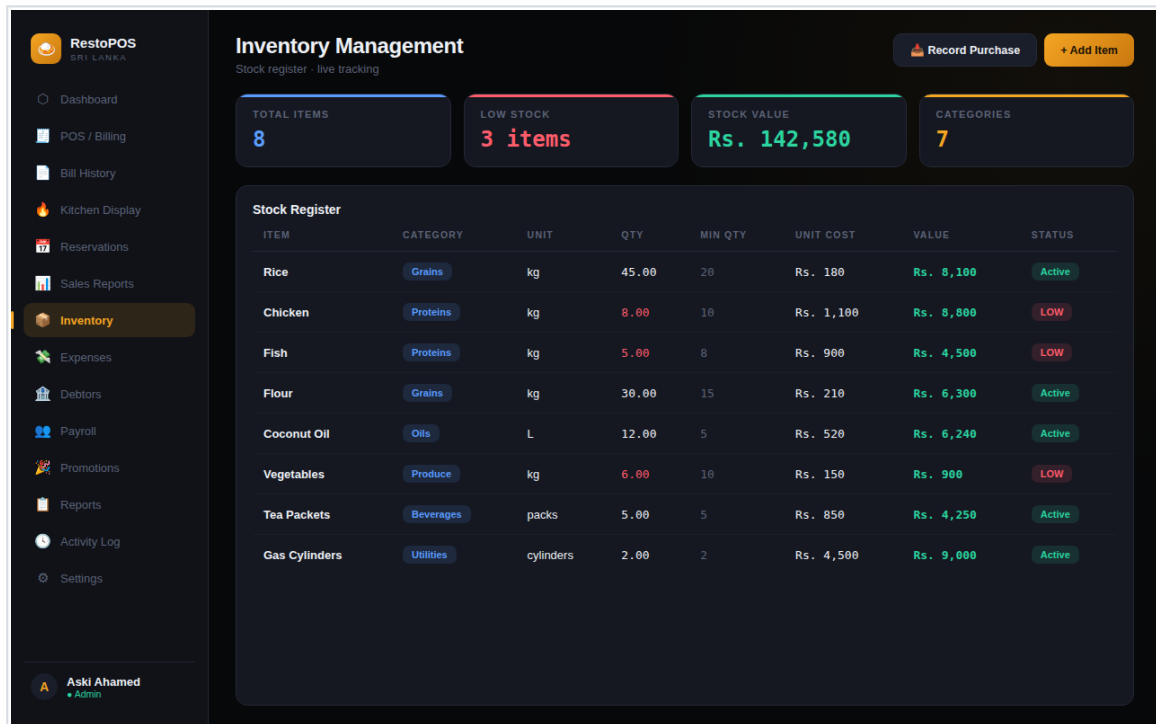


Figure 8: Inventory register showing stock levels, unit costs, and low-stock indicators

3.9 Promotions Engine

The Promotions module supports three distinct discount mechanisms: a fixed percentage reduction, a fixed Rupee-amount reduction, and a "Buy X, Get Y Free" quantity-based promotion. The latter required custom calculation logic that groups cart items by product and determines, for each qualifying group, how many units should be discounted to zero cost.

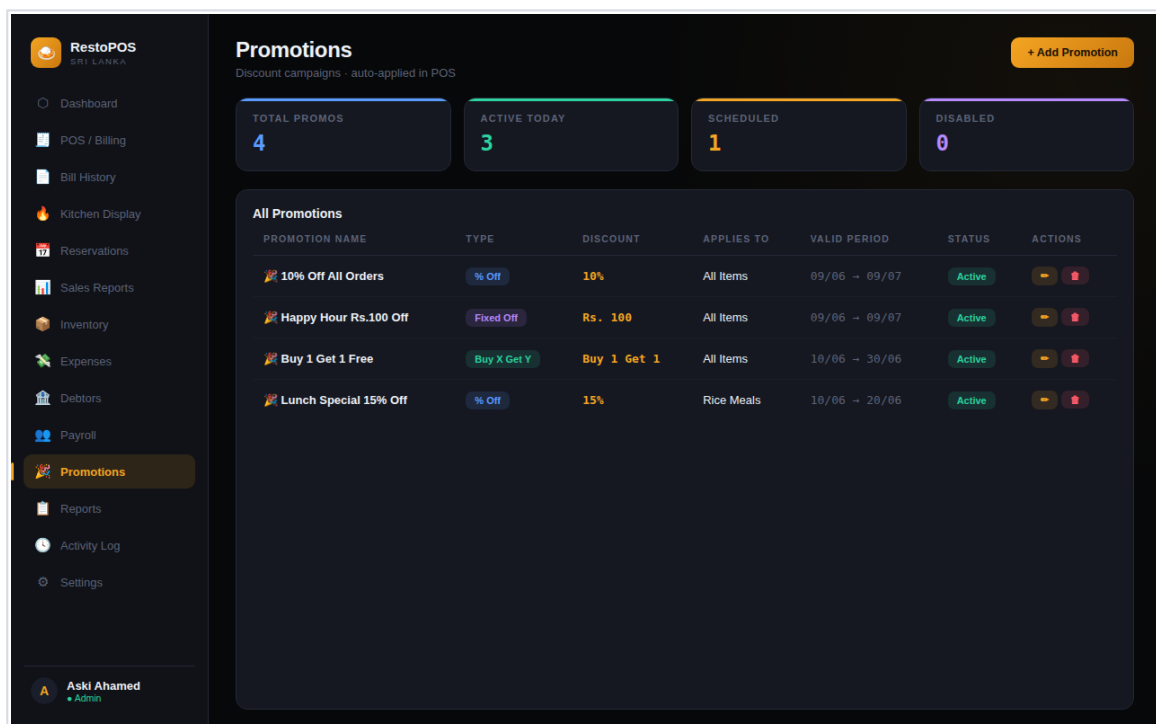


Figure 9: Promotions management interface listing all active and scheduled campaigns

Active promotions, filtered by their configured validity window, surface automatically within the POS interface as selectable cards. Applied promotions are persisted against the originating bill, enabling subsequent reporting on promotional cost and usage frequency (see Section 3.12).

3.10 Payroll (EPF/ETF Compliant)

The Payroll module implements Sri Lanka's statutory retirement contribution scheme: an 8% Employees' Provident Fund (EPF) deduction from employee gross pay, a 12% EPF employer contribution, and a 3% Employees' Trust Fund (ETF) employer contribution — calculated automatically for EPF-applicable staff and waived entirely for staff marked as non-applicable (e.g. part-time or probationary employees, per employer policy).

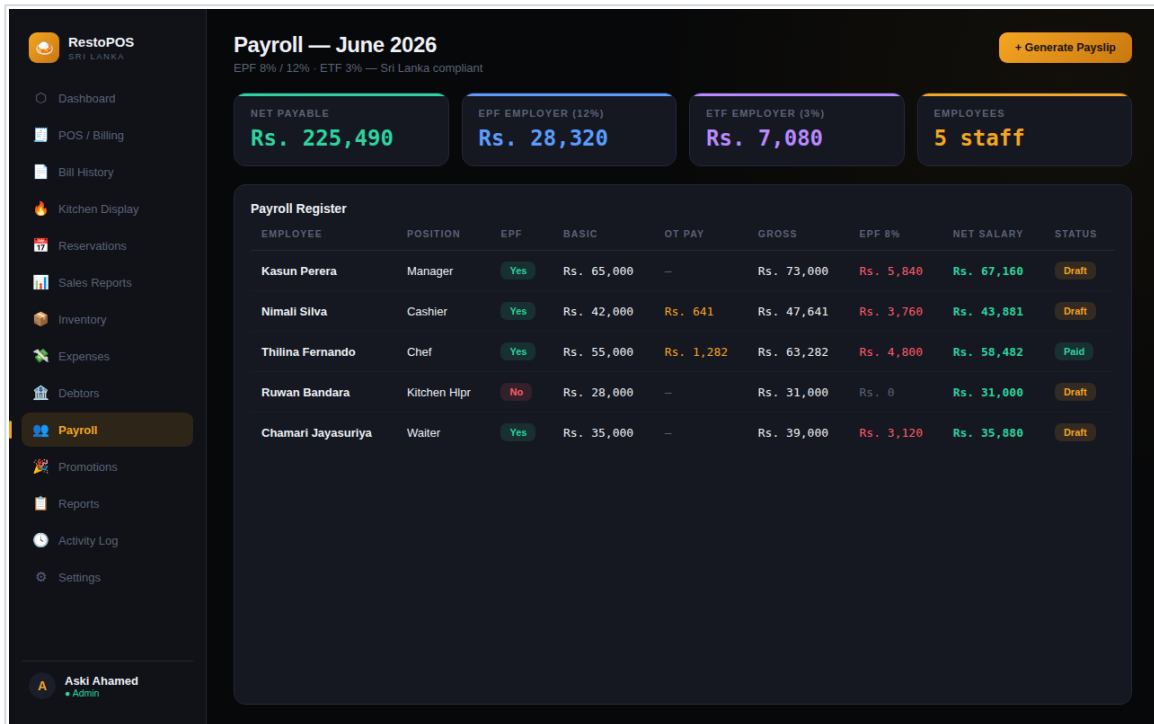


Figure 10: Payroll register showing automatic EPF/ETF calculation per employee

Overtime is calculated using a standard nine-hour shift baseline measured from each employee's individually recorded clock-in time, rather than a fixed organisation-wide cut-off. Hours worked beyond this nine-hour threshold accrue at a flat overtime rate, configured at Rs. 100 per hour. This design choice — calculating the threshold relative to each employee's actual start time rather than a universal clock time — was adopted to correctly handle staff with staggered shift starts.

3.11 Employees & Attendance

This module maintains the employee register and daily attendance records. Attendance may be marked individually or in bulk for the entire active staff roster in a single operation. Overtime hours, where applicable, are auto-calculated from recorded Time In and Time Out values and flow directly into the Payroll module without requiring duplicate manual entry.

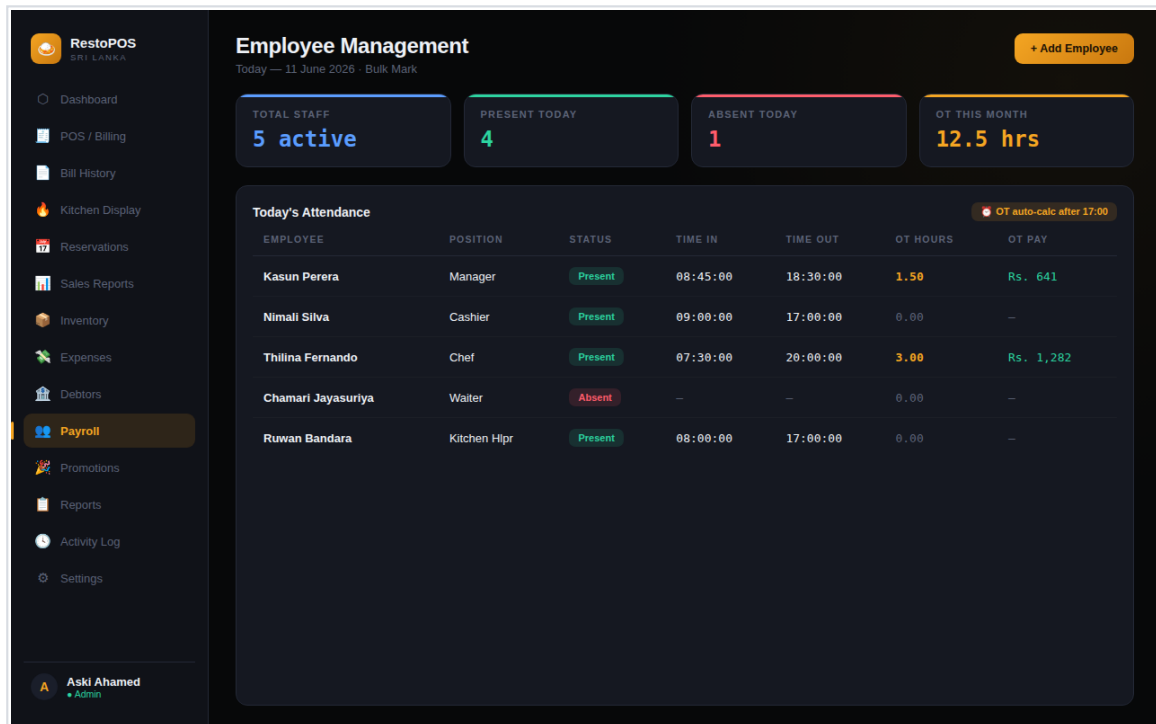


Figure 11: Daily attendance register with automatic overtime calculation

3.12 Financial Reports Suite

A consolidated Reports module presents eight distinct report types — Profit & Loss, Expenses, Cash Flow, Debtors, Inventory, Payroll, Attendance, and Promotions — each independently filterable by date range and exportable in two formats: a browser-printable PDF layout, and a UTF-8 encoded CSV file compatible with Microsoft Excel.

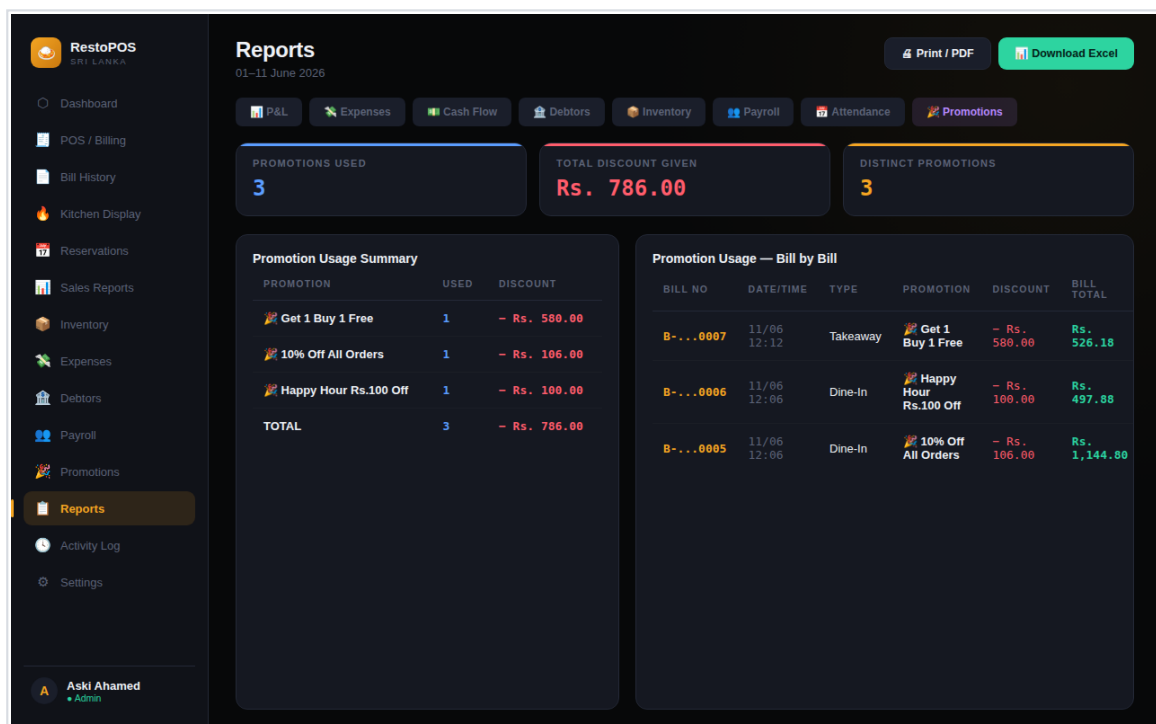


Figure 12: Promotions usage report, showing per-campaign discount totals and bill-level detail

Access to the Payroll and Attendance report tabs is restricted to Administrator and Manager roles; Cashiers are denied access at the navigation, page-request, and CSV/PDF export layers, ensuring payroll data confidentiality is enforced consistently regardless of access method.

3.13 Activity Log

To support accountability and operational audit requirements, the system maintains a comprehensive Activity Log recording significant user actions — including login and logout events, bill settlement, voiding, and deletion, and reservation creation and modification. Each log entry captures the acting user, the action performed, the affected module, a human-readable description, a timestamp, and the originating IP address.

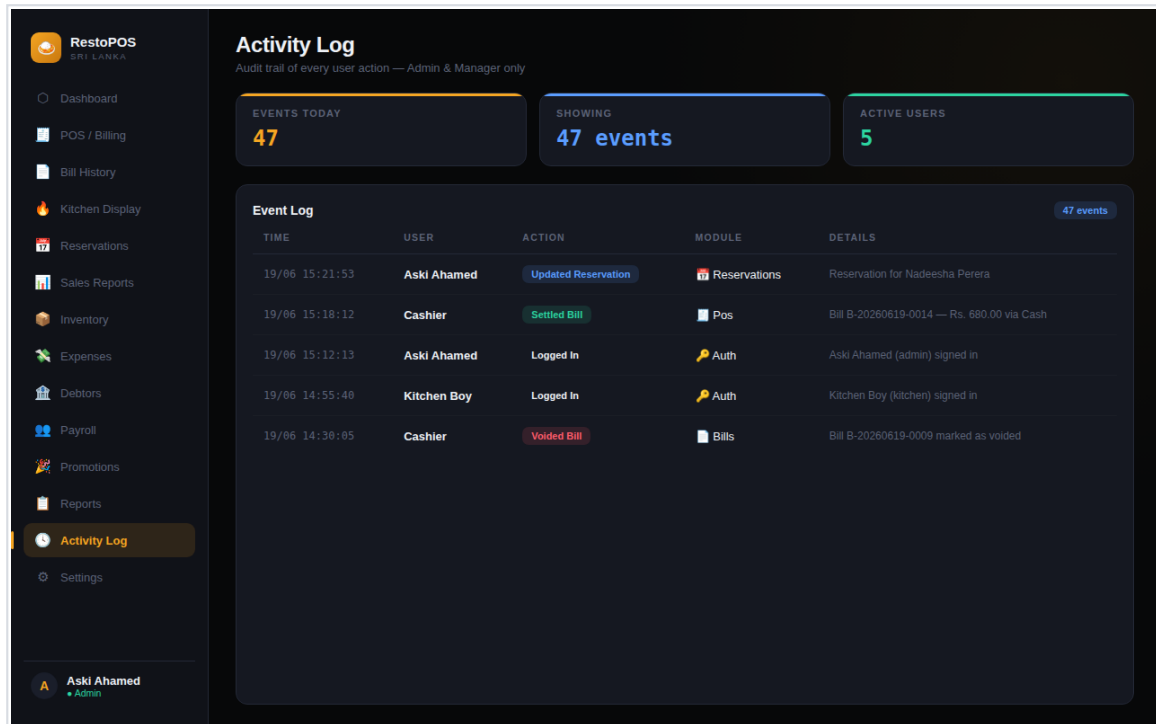


Figure 13: Activity Log showing a filtered audit trail of user actions

3.14 Settings & User Management

The Settings module, restricted to the Administrator role, centralises business configuration — business name, address, contact details, service charge percentage, tax/VAT rate, and third-party delivery platform commission rates for Uber Eats and PickMe. The same module hosts System Users management, where administrators create, edit, deactivate, or delete staff login accounts and assign their role.

CHAPTER 4

Database Design

The system persists data across twenty-two relational tables in a MySQL database, designed around a normalised schema with foreign-key relationships linking transactional records (bills, bill items, attendance, payroll) to their respective master records (menu items, employees, inventory items).

Core Entity Groups

Identity & Access	users, settings
Menu & Billing	menu_categories, menu_items, bills, bill_items, bill_promotions
Promotions	promotions
Inventory	inventory_categories, inventory_items, stock_purchases
Human Resources	employees, attendance, payroll
Customer Relations	debtors, debtor_payments, reservations
Finance	expense_categories, expenses, bank_accounts, bank_transactions
Operations & Audit	activity_log

Notable Design Decisions

- **kitchen_status** column on `bill_items`: rather than introducing a separate kitchen-order table, kitchen workflow state is tracked directly on the existing `bill_items` record, simplifying the join logic required by the Kitchen Display System.
- **role** column on `users`: implemented as `VARCHAR` rather than a fixed `ENUM` list. This decision was made after an early implementation using `ENUM` caused new role values to be silently rejected by MySQL when not pre-declared in the column definition — a defect documented in Chapter 5.
- Timestamps throughout the schema default to the application server's local timezone (Asia/Colombo) rather than relying on the database server's default UTC, preventing date-boundary discrepancies in daily reporting.
- Bill totals are persisted as calculated, immutable values at the point of settlement, rather than recalculated on each view — preserving historical accuracy even if menu prices or tax rates change subsequently.

CHAPTER 5

Testing & Validation

Development of RestoPOS Sri Lanka followed an iterative cycle of feature implementation, user acceptance testing against the live deployed instance, and defect resolution. Several notable defects were identified during real-world usage and resolved as follows:

| Timezone-related date filtering

Bills created near end-of-day were not appearing under "Today" filters. Root cause: MySQL CURDATE() resolved against the database server's UTC clock while PHP-side filtering used Asia/Colombo local time. Resolved by enforcing `date_default_timezone_set('Asia/Colombo')` and replacing all CURDATE() usage with PHP-generated date values passed as bound parameters.

| Conflicting JavaScript function declarations

The Cash Given "Change" amount displayed an incorrect value after a promotional discount was introduced mid-development. Root cause: two independently defined functions of the same name existed in separate script files; whichever loaded last silently overrode the other depending on script tag order. Resolved by consolidating all bill-calculation logic into a single authoritative function, with both script files explicitly deferring to it regardless of load order.

| Silent ENUM rejection on role assignment

Newly introduced user roles (Kitchen Staff) failed to persist when assigned via the user-management interface, with no error surfaced to the administrator. Root cause: the role column was defined as a MySQL ENUM with a fixed, pre-declared value list; MySQL silently substitutes an empty string for any value outside that list rather than raising an exception. Resolved by migrating the column to VARCHAR with application-layer validation.

| Buy X Get Y promotion discount calculation

The "Buy 1, Get 1 Free" promotion type initially applied no discount under any cart configuration. Root cause: the discount-calculation function lacked any implementation branch for the `buy_x_get_y` promotion type. Resolved by implementing per-item quantity grouping logic that identifies qualifying free units for each distinct product in the cart.

CHAPTER 6

Conclusion & Future Work

6.1 Conclusion

RestoPOS Sri Lanka successfully delivers an integrated restaurant management platform addressing the operational requirements outlined in Chapter 1. The system has been deployed and validated in a live operational context, processing real billing transactions, managing live inventory and staff attendance data, and coordinating kitchen workflow through the Kitchen Display System. The role-based access control model has proven effective in restricting sensitive payroll and financial data from operational staff while preserving the workflow efficiency of front-of-house and kitchen personnel.

6.2 Future Work

Several enhancements have been identified as candidates for subsequent development phases:

1. A public-facing online menu, accessible via QR code, allowing customers to browse offerings on their personal devices ahead of or during their visit.
2. Customer loyalty and rewards point accumulation, integrated with the existing Promotions engine.
3. Multi-branch support, enabling a single administrative account to oversee inventory, staffing, and sales reporting across multiple restaurant locations.
4. SMS or WhatsApp-based reservation confirmation and reminder notifications.
5. Migration of password storage from plain text to a salted cryptographic hash, strengthening account security ahead of any production deployment handling sensitive customer payment data.

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